12TH BIOLOGY · CHAPTER 14 · FULL CHAPTER MCQS

Human Urinary System

78 topic-wise MCQs from the complete chapter with answers and reasons

14.1 HUMAN URINARY SYSTEM

1. The process by which an organism eliminates its metabolic wastes is called:

(a) Egestion (b) Osmosis
(c) Secretion (d) Excretion

Answer: (d) Excretion — This is the definition of excretion.

2. The urinary system keeps the pH of blood around:

(a) 7.4 (b) 7.0
(c) 6.4 (d) 8.4

Answer: (a) 7.4 — Blood is kept slightly basic, around pH 7.4.

3. Urea is a nitrogenous waste of protein breakdown in the:

(a) Kidney (b) Urinary bladder
(c) Liver (d) Pancreas

Answer: (c) Liver — Urea is formed in the liver and filtered out by the kidneys.

4. The human urinary system has how many ureters?

(a) One (b) Two
(c) Three (d) Four

Answer: (b) Two — Two kidneys, two ureters, one urinary bladder and one urethra.

5. Kidneys are located between which vertebrae?

(a) 12th thoracic and 3rd lumbar (b) 1st thoracic and 12th thoracic
(c) 3rd lumbar and 5th lumbar (d) 7th cervical and 1st thoracic

Answer: (a) 12th thoracic and 3rd lumbar — They lie on each side of the vertebral column at this level.

6. The right kidney is slightly lower than the left because of the position of the:

(a) Stomach (b) Heart
(c) Liver (d) Spleen

Answer: (c) Liver — The liver lies above the right kidney.

7. The upper parts of the kidneys are partially protected by the:

(a) 9th and 10th ribs (b) Pelvic bones
(c) 1st and 2nd ribs (d) 11th and 12th ribs

Answer: (d) 11th and 12th ribs — The 11th and 12th ribs protect them.

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8. The length of a ureter is about:

(a) 5 cm (b) 25 cm
(c) 10 cm (d) 50 cm

Answer: (b) 25 cm — Ureters are about 25 cm long tubes.

9. Which organ serves as the urine reservoir?

(a) Kidney (b) Ureter
(c) Urinary bladder (d) Urethra

Answer: (c) Urinary bladder — It is a hollow, distensible muscular organ in the pelvic cavity.

10. The tube that carries urine from the urinary bladder to the outside is the:

(a) Urethra (b) Ureter
(c) Collecting duct (d) Renal pelvis

Answer: (a) Urethra — The urethra carries urine out of the body.

11. The gland present on the top of each kidney is the:

(a) Thyroid gland (b) Pituitary gland
(c) Pancreas (d) Adrenal gland

Answer: (d) Adrenal gland — Each kidney has an adrenal gland on its top.

12. The depressed area on the concave side of the kidney is the:

(a) Renal pelvis (b) Renal hilum
(c) Renal pyramid (d) Renal capsule

Answer: (b) Renal hilum — The renal artery enters and the renal vein and ureter leave here.

13. The renal cortex makes up about _____ of the kidney's mass.

(a) One-third (b) Two-thirds
(c) One-half (d) One-fifth

Answer: (a) One-third — The medulla makes up two-thirds.

14. The cone-shaped structures of the renal medulla are called:

(a) Renal corpuscles (b) Glomeruli
(c) Renal capsules (d) Renal pyramids

Answer: (d) Renal pyramids — The medulla consists of a number of renal pyramids.

15. The funnel-shaped structure in the centre of the kidney is the:

(a) Renal hilum (b) Renal cortex
(c) Renal pelvis (d) Bowman's capsule

Answer: (c) Renal pelvis — Urine is collected in the renal pelvis.

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16. Each kidney contains about how many nephrons?

(a) One thousand (b) One million
(c) One hundred thousand (d) One billion

Answer: (b) One million — About one million nephrons are present in each kidney.

17. The renal corpuscle consists of:

(a) Glomerulus and Bowman’s capsule (b) Loop of Henle and collecting duct
(c) Proximal and distal tubules (d) Afferent and efferent arterioles

Answer: (a) Glomerulus and Bowman’s capsule — These two structures make the renal corpuscle.

18. The cup-shaped, double-walled sac of the nephron is the:

(a) Glomerulus (b) Loop of Henle
(c) Renal pelvis (d) Bowman’s capsule

Answer: (d) Bowman’s capsule — It surrounds the glomerulus.

19. The U-shaped part of the renal tubule is the:

(a) Proximal convoluted tubule (b) Distal convoluted tubule
(c) Loop of Henle (d) Collecting duct

Answer: (c) Loop of Henle — It has a descending and an ascending limb.

20. Cortical nephrons make about _____ of the nephrons.

(a) 15% (b) 85%
(c) 50% (d) 100%

Answer: (b) 85% — The majority (about 85%) are cortical nephrons.

21. Long loops of Henle that extend deep into the medulla are found in:

(a) Cortical nephrons (b) Collecting ducts
(c) All nephrons (d) Juxtamedullary nephrons

Answer: (d) Juxtamedullary nephrons — Cortical nephrons have short loops.

22. Collecting ducts merge to form larger ducts called:

(a) Papillary ducts (b) Ureters
(c) Renal pyramids (d) Vasa recta

Answer: (a) Papillary ducts — Papillary ducts discharge into the renal pelvis.

23. The glomerular capillaries drain into the:

(a) Renal venule (b) Afferent arteriole
(c) Efferent arteriole (d) Renal vein

Answer: (c) Efferent arteriole — Unlike other capillary beds, they drain into an arteriole.

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24. The capillaries that surround the proximal and distal convoluted tubules are:

(a) Glomerular capillaries (b) Peritubular capillaries
(c) Vasa recta (d) Renal venules

Answer: (b) Peritubular capillaries — They arise from the efferent arteriole.

25. Efferent arterioles of juxtamedullary nephrons give rise to:

(a) Peritubular venules (b) Afferent arterioles
(c) Interlobular arteries (d) Vasa recta

Answer: (d) Vasa recta — Vasa recta run parallel to the loops of Henle.

26. The renal vein drains into the:

(a) Aorta (b) Superior vena cava
(c) Inferior vena cava (d) Hepatic vein

Answer: (c) Inferior vena cava — It returns the filtered blood towards the heart.

27. Kidneys receive approximately what share of the total cardiac output?

(a) 20–25% (b) 5–10%
(c) 1–2% (d) 50–60%

Answer: (a) 20–25% — Kidneys are highly vascularized organs.

28. The first step in the formation of urine is:

(a) Tubular secretion (b) Glomerular filtration
(c) Selective reabsorption (d) Elimination

Answer: (b) Glomerular filtration — It is also called ultrafiltration.

29. Blood in the glomerulus is under high pressure because the efferent arteriole is:

(a) Narrower than the afferent (b) Wider than the afferent
(c) Equal to the afferent (d) Absent

Answer: (a) Narrower than the afferent — Blood leaves less easily than it enters.

30. About what fraction of the fluid portion of blood filters into the Bowman's capsule?

(a) One-half (b) One-third
(c) One-tenth (d) One-fifth

Answer: (d) One-fifth — About 20% forms the glomerular filtrate.

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31. Which of these is NOT present in the glomerular filtrate of a healthy person?

(a) Glucose (b) Urea
(c) Albumin (d) Amino acids

Answer: (c) Albumin — Plasma proteins are too large to pass through the capillary walls.

32. Most reabsorption occurs in the:

(a) Loop of Henle (b) Proximal convoluted tubule
(c) Distal convoluted tubule (d) Collecting duct

Answer: (b) Proximal convoluted tubule — About 75% of the water is reabsorbed here.

33. In the proximal tubule, water returns to the capillaries by:

(a) Active transport (b) Filtration
(c) Osmosis (d) Secretion

Answer: (c) Osmosis — Glucose, amino acids and salts return by active transport.

34. The final site for water reabsorption is the:

(a) Collecting duct (b) Proximal tubule
(c) Loop of Henle (d) Bowman's capsule

Answer: (a) Collecting duct — The collecting duct is the final site.

35. Tubular secretion takes place in the:

(a) Glomerulus (b) Distal convoluted tubule
(c) Bowman's capsule (d) Renal pelvis

Answer: (b) Distal convoluted tubule — Its epithelial cells secrete materials into the filtrate.

36. Which of these is secreted into the filtrate during tubular secretion?

(a) Glucose (b) Albumin
(c) Amino acids (d) Hydrogen ions

Answer: (d) Hydrogen ions — H^+, ammonium, K^+, toxins and drug residues are secreted.

37. A normal adult eliminates how much urine a day?

(a) 100–200 mL (b) 5–6 L
(c) 1–2 L (d) 10 L

Answer: (c) 1–2 L — The amount is variable and depends on several factors.

38. A diet with high salt:

(a) Increases urine production (b) Decreases urine production
(c) Stops urine production (d) Has no effect

Answer: (b) Decreases urine production — High salt triggers water retention.

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14.2 REGULATION OF THE URINE CONCENTRATION

39. The descending limb of the loop of Henle is:

(a) Permeable to water but not to salts (b) Permeable to salts but not to water
(c) Impermeable to both (d) Permeable to proteins

Answer: (a) Permeable to water but not to salts — Water is reabsorbed from it.

40. The ascending limb of the loop of Henle:

(a) Reabsorbs water
(b) Secretes urea
(c) Is impermeable to water and actively transports salts out
(d) Filters blood

Answer: (c) Is impermeable to water and actively transports salts out — This keeps the surrounding fluid salty.

41. The high salt concentration of the fluid around the nephron is maintained by:

(a) Glomerular filtration (b) Counter-current mechanism
(c) Tubular secretion (d) Dialysis

Answer: (b) Counter-current mechanism — It allows water reabsorption from the loop and collecting duct.

42. Water leaving the collecting duct enters the blood via the:

(a) Glomerulus (b) Afferent arteriole
(c) Renal artery (d) Vasa recta

Answer: (d) Vasa recta — Vasa recta run in the medulla.

43. ADH is produced by the:

(a) Adrenal cortex (b) Hypothalamus
(c) Kidney (d) Thyroid

Answer: (b) Hypothalamus — It is secreted by the posterior pituitary gland.

44. ADH is secreted (released) by the:

(a) Anterior pituitary (b) Adrenal cortex
(c) Posterior pituitary (d) Adrenal medulla

Answer: (c) Posterior pituitary — It is produced by the hypothalamus.

45. Osmoreceptors that detect the salt concentration of blood are present in the:

(a) Hypothalamus (b) Kidney
(c) Adrenal gland (d) Urinary bladder

Answer: (a) Hypothalamus — They trigger thirst and ADH release.

46. ADH makes the walls of the collecting ducts:

(a) Less permeable to water (b) Impermeable to salts
(c) Permeable to proteins (d) More permeable to water

Answer: (d) More permeable to water — So more water is reabsorbed.

47. In the absence of ADH, the urine is:

(a) Concentrated and small in volume (b) Absent
(c) Dilute and large in volume (d) Rich in glucose

Answer: (c) Dilute and large in volume — Less water is reabsorbed.

48. Lack of ADH due to pituitary damage causes:

(a) Diabetes insipidus (b) Diabetes mellitus
(c) Hyperkalaemia (d) Kidney stones

Answer: (a) Diabetes insipidus — The patient excretes a large volume of dilute urine.

49. Aldosterone is produced by the:

(a) Posterior pituitary (b) Adrenal medulla
(c) Hypothalamus (d) Adrenal cortex

Answer: (d) Adrenal cortex — The adrenal cortex is the outer layer of the adrenal gland.

50. Aldosterone increases the reabsorption of:

(a) Potassium ions (b) Sodium ions
(c) Hydrogen ions (d) Urea

Answer: (b) Sodium ions — Water follows sodium into the blood.

51. Aldosterone is secreted when:

(a) Blood pressure drops or blood sodium is too low
(b) Blood pressure rises
(c) Blood glucose rises
(d) Urine is dilute

Answer: (a) Blood pressure drops or blood sodium is too low — It restores sodium and water in the blood.

52. Aldosterone promotes the excretion of which ions in urine?

(a) Sodium (b) Chloride
(c) Calcium (d) Potassium

Answer: (d) Potassium — This helps to prevent hyperkalaemia.

53. Hyperkalaemia is a high blood level of:

(a) Sodium (b) Potassium
(c) Calcium (d) Uric acid

Answer: (b) Potassium — It can lead to dangerous cardiac arrhythmias.

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54. Glomerular capillaries are designed for:

(a) Reabsorption (b) Secretion
(c) Filtration (d) Storage

Answer: (c) Filtration — They operate under high blood pressure.

55. Peritubular capillaries operate under:

(a) Low hydrostatic pressure (b) High blood pressure
(c) No pressure (d) Osmotic pressure only

Answer: (a) Low hydrostatic pressure — Their role is reabsorption and secretion.

14.3 DISORDERS OF URINARY TRACT

56. Urinary tract infections are most commonly caused by:

(a) Clostridium (b) Streptococcus
(c) Rhizobium (d) Escherichia coli

Answer: (d) Escherichia coli — The bacteria enter through the urethra.

57. UTIs are treated with:

(a) Antibiotics (b) Lithotripsy
(c) Dialysis (d) Immunosuppressants

Answer: (a) Antibiotics — Antibiotics kill the causative bacteria.

58. About 70% of kidney stone patients have stones of:

(a) Uric acid (b) Cysteine
(c) Calcium oxalate (d) Calcium phosphate

Answer: (c) Calcium oxalate — They form due to hypercalciuria and hyperoxaluria.

59. About 10% of kidney stone patients have stones of:

(a) Calcium oxalate (b) Uric acid
(c) Cysteine (d) Phosphate

Answer: (b) Uric acid — Hyperuricemia is their cause.

60. Hypercalciuria means:

(a) High calcium in blood (b) High oxalate in urine
(c) Low calcium in urine (d) High calcium in urine

Answer: (d) High calcium in urine — It leads to calcium oxalate stones.

61. Increased amount of uric acid in blood is called:

(a) Hyperoxaluria (b) Hypercalciuria
(c) Hyperuricemia (d) Hyperkalaemia

Answer: (c) Hyperuricemia — It causes uric acid stones.

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62. When urine is alkaline, the kidney stone is of:

(a) Calcium oxalate (b) Calcium phosphate
(c) Uric acid (d) Cysteine

Answer: (b) Calcium phosphate — Acidic urine indicates calcium oxalate; persistently acidic, uric acid.

63. Breaking of kidney stones is called:

(a) Lithotripsy (b) Dialysis
(c) Transplant (d) Filtration

Answer: (a) Lithotripsy — ECSWL and PNL are its two methods.

64. In ECSWL, shockwaves are generated by an instrument called:

(a) Nephroscope (b) Lithotripter
(c) Dialyzer (d) Catheter

Answer: (b) Lithotripter — It works from outside the body.

65. The small camera used in PNL is called:

(a) Nephroscope (b) Lithotripter
(c) Dialyzer (d) Endoscope

Answer: (a) Nephroscope — It is inserted through a tube from the patient's back.

66. PNL is used for:

(a) Small stones (b) UTIs
(c) Kidney failure (d) Larger stones

Answer: (d) Larger stones — The pieces are pulled out with special equipment.

67. Chronic kidney failure is:

(a) Sudden and reversible (b) Sudden and irreversible
(c) Slow and irreversible (d) Always due to stones

Answer: (c) Slow and irreversible — It develops over a longer period.

68. Which of these may cause acute kidney failure?

(a) Glucose (b) Vitamins
(c) Dialysate (d) NSAIDs

Answer: (d) NSAIDs — Certain medications such as NSAIDs can cause it.

69. The semipermeable membrane of the dialyzer is made of:

(a) Chitin (b) Cellulose
(c) Keratin (d) Protein

Answer: (b) Cellulose — The dialyzer is a type of filter.

70. The special fluid used in dialysis is called:

(a) Dialysate (b) Filtrate
(c) Plasma (d) Lymph

Answer: (a) Dialysate — Wastes, water and salts move from blood into it.

71. In peritoneal dialysis, how much dialysate is infused into the abdomen?

(a) 0.5 L (b) 5 L
(c) 1.5 to 2 L (d) 10 L

Answer: (c) 1.5 to 2 L — It is left for 4 to 6 hours.

72. In peritoneal dialysis, the dialysate is left in the abdomen for:

(a) 4 to 6 minutes (b) 4 to 6 hours
(c) 4 to 6 days (d) 24 hours

Answer: (b) 4 to 6 hours — Then it is drained and discarded.

73. A common complication of haemodialysis is:

(a) Hypotension (b) Hypertension
(c) Hyperkalaemia (d) Kidney stones

Answer: (a) Hypotension — Low blood pressure is common.

74. Kidney transplant is offered to patients with:

(a) UTIs (b) Small kidney stones
(c) Diabetes insipidus (d) End-stage kidney failure

Answer: (d) End-stage kidney failure — The non-functioning kidney is replaced with a healthy one.

75. The best donor of a kidney is a:

(a) Friend (b) Co-worker
(c) Related donor (d) Any stranger

Answer: (c) Related donor — A parent, sibling or child is genetically related.

76. Organ rejection after transplant is caused by the recipient's:

(a) Urinary system (b) Immune system
(c) Nervous system (d) Endocrine system

Answer: (b) Immune system — It recognizes the kidney as a foreign object.

77. Pakistan's first kidney transplant centre (1985) was established at:

(a) PKLI, Lahore (b) Shifa, Islamabad
(c) AFIU, Rawalpindi (d) SIUT, Karachi

Answer: (d) SIUT, Karachi — Sindh Institute of Urology and Transplantation.

78. PKLI&RC, Lahore was established in:

(a) 1985 (b) 2005
(c) 2017 (d) 2025

Answer: (c) 2017 — It is a tertiary-care hospital for kidney and liver diseases.

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